

26 Null Hypothesis Testing

- Statisticians view conjectures about a population as competing *hypotheses*. These hypotheses usually constitute claims about the values of certain parameters.
- A statistical **(null) hypothesis test** consists of using data to make a decision regarding two explanations (hypotheses).
- The **null hypothesis**, denoted H_0 , represents the “status quo/no difference/no effect/no association” hypothesis.
 - If the null hypothesis is true then any observed difference/effect/association is just due to “random chance” resulting from natural sampling variability
- The **alternative hypothesis**, denoted H_a (or H_1), typically corresponds to what the researchers are hoping to gather evidence to support, and thus is a statement in line with the research conjecture
- A statistical hypothesis test assesses if the data provide evidence to rule out the “random chance” explanation (which corresponds to the null hypothesis) in favor of the explanation provided by the research conjecture (alternative hypothesis).
- The null hypothesis is often of the form

$$H_0 : \text{parameter} = \text{hypothesized value}$$

where the hypothesized value is a specific number determined by the research question (often 0 for “no difference”).

- The alternative hypotheses is often of one of the following forms

Case 1a: $H_a : \text{parameter} > \text{hypothesized value}$

Case 1b: $H_a : \text{parameter} < \text{hypothesized value}$

Case 2: $H_a : \text{parameter} \neq \text{hypothesized value}$ (“two-sided”)

Which of the three cases above is appropriate is determined by the research conjecture. When in doubt, use a two-sided ($\neq$) alternative.

Example 26.1. Did the New England Patriots deliberately deflate footballs in an NFL playoff game against the Indianapolis Colts on January 18, 2015? The following table¹ summarizes the decrease in air pressure (psi), from pre-game to halftime, for the footballs from each team that were checked at halftime by officials.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

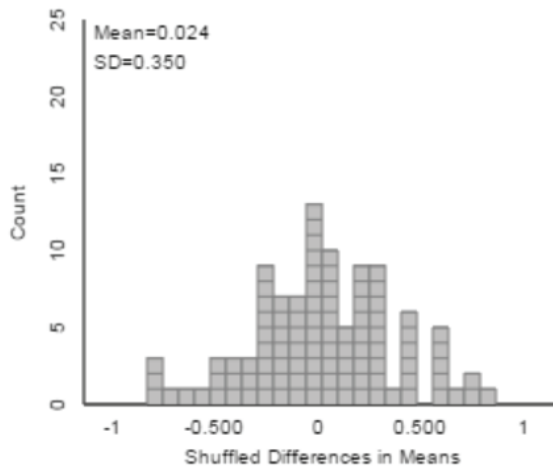
Deflation														
Team	(psi)											Size	Mean	SD
Patriots	1.00	1.65	1.35	1.80	1.40	0.90	0.65	1.40	1.55	2.00	1.60	11	1.391	0.402
Colts	0.30	0.25	0.50	0.45								4	0.375	0.119

1. Compute the observed difference in sample means.
2. If there were no difference between the Patriots and the Colts footballs, what would you expect the difference in sample means to be? Is the fact that we observed another value necessarily evidence that the Patriots footballs tended to deflate more than the Colts?
3. Suppose we want to assess if it is plausible that the Patriots didn't interfere with the footballs, and the observed difference in sample means is just due to chance variability. Specify how the "null distribution" of the statistic could be simulated with cards. Note: this is NOT bootstrapping; how is it different?
4. Use cards to perform one repetition of the above simulation, record the results here, and compute the appropriate difference in means.

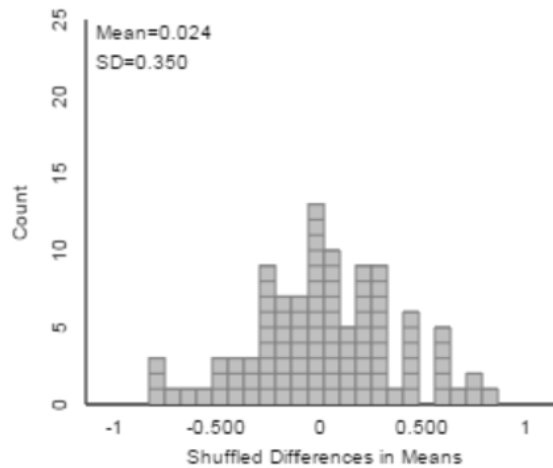
Team	Deflation (psi)										Size	Mean	SD
Patriots											11		
Colts						X	X	X	X	X	X	4	

5. The following plot displays results from an [applet](#) used to run 10000 repetitions of the simulation (the plot on the left displays the results from just the first 100 repetitions). Estimate the probability of observing a difference in sample means as extreme as the observed difference just by chance if there were no real difference between the Patriots and Colts footballs.

Total Shuffles = 100



Total Shuffles = 100



Count Samples

(a) First 100 repetitions

Count Samples

(b) 10000 repetitions

Figure 26.1: Permutation simulation of null distribution in Example 26.1.

6. Is it plausible that the observed difference in means is just due to random chance? To support your answer, interpret the probability from the previous part in context.

- To assess the plausibility of the “random chance” explanation for the observed result — corresponding to the null hypothesis being true — we need to know the pattern of sampling variability of the statistic that would be expected just due to random chance if the null hypothesis were true. This is called the **null distribution** of the statistic.
- If the actual observed value of the statistic is not consistent with this pattern, then we have convincing evidence that the results did not occur by random chance alone, and we have evidence to reject the null hypothesis
- The (simulated) null distribution provides a picture of what typical values of the statistic would be expected just by random chance alone if the null hypothesis were true.
- The question is then: how surprising is the result we actually observed from the perspective of the null distribution? If the result would be very surprising, then “random chance” is not a plausible explanation for the observed result.

- This gives us evidence to rule out a “random chance” explanation like in favor of a “research conjecture” explanation.
- In other words, the more surprising the observed result would be just due to random chance, the stronger the evidence to **reject the null hypothesis** in favor of the alternative hypothesis.
- We quantify “how surprising” with a p-value.
 - The **p-value** is the probability of observing a value at least as extreme as the value actually observed, when the null hypothesis is true.
 - The p-value can be estimated by the proportion (out of the total number of repetitions) of values in the simulated null distribution that are *at least as extreme* as the actual observed value of the statistic.
 - * Which tail(s) constitutes “at least as extreme” is determined by the form of the *alternative hypothesis*
- The smaller the p-value — that is, the closer the p-value is to 0 — the less plausible it is that the observed result occurred just by random chance alone.
- Thus, **the smaller the p-value the stronger the evidence to reject the null hypothesis in favor of the alternative hypothesis.**
- If the observed results provide strong evidence that the data did not arise by random chance alone — that is, when the p-value is small — then there is evidence of a **statistical difference**.
- How small is a small p-value? You should think of a sliding scale: the closer the p-value gets to 0 the stronger the evidence to reject the null hypothesis in favor of the alternative. Some guidelines are below, but these should *not* be followed too strictly (especially when performing many hypothesis tests simultaneously):
- Do NOT rely strictly on arbitrary thresholds like $p\text{-value} < 0.05$.

p-value	Evidence to reject the null hypothesis?
Above 0.10	Little to no evidence
Between 0.01 and 0.10	Weak evidence
Between 0.001 and 0.01	Moderate evidence
Between 0.0001 and 0.001	Strong evidence
Less than 0.0001	Very strong evidence

Example 26.2. Carrying a heavy backpack can cause chronic shoulder, neck, and back pain. A [common recommendation](#) is that a daily backpack should not weigh more than 10 percent of the wearer’s body weight. Do Cal Poly students follow this recommendation? That is, do the backpacks of Cal Poly students weigh, on average, less than 10% of their bodyweight? In a random sample of 100 Cal Poly students the sample mean backpack weight as percent of bodyweight is 7.71 percent with sample standard deviation 0.366 percent.

1. Identify the main parameter of interest with both words and an appropriate symbol.
2. Translate the research question into a hypothesis testing problem by specifying, with both words and appropriate symbols, the null hypothesis and the alternative hypothesis.
3. Identify the null distribution of sample means. (You might need to make some assumptions.)
4. Compute the p-value.
5. Interpret the p-value in context.
6. Does the p-value provide evidence that backpacks of CalPoly students weigh, on average, less than 10% of their bodyweight? Explain.
7. Does the p-value provide evidence that backpacks of CalPoly students weigh, on average, *much* less than 10% of their bodyweight? Explain.

8. How could we address the question in the previous part?

9. Do Cal Poly students follow the 10% recommendation? Is the analysis above the only way to address this question? Can you suggest another?

One-sample t test

- Situation: question concerns a single numerical variable of interest
- Parameter: μ is the population mean (i.e. the mean value of the variable over all units in the population)
- Goal: assess the plausibility of a claim about μ regarding a hypothesized value μ_0
- The null hypothesis is $H_0 : \mu = \mu_0$
- The alternative hypothesis is one of $H_a : \mu > \mu_0$, $H_a : \mu < \mu_0$, or $H_a : \mu \neq \mu_0$
 - When in doubt, use a two-sided alternative and a two-tailed p-value.
- Statistics: $\bar{x}$ is the sample mean; s is the sample standard deviation
- Test statistic: the standardized value of $\bar{x}$, where
 - the mean is computed assuming the null hypothesis, $H_0 : \mu = \mu_0$, is true, and
 - the unknown SD ($\sigma/\sqrt{n}$) is estimated with the standard error $s/\sqrt{n}$
$$t = \frac{\text{observed} - \text{hypothesized}}{\text{SE}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
- The p-value is the probability of observing a value at least as extreme as t , in the tail direction indicated by the alternative hypothesis, for the t -distribution with $n-1$ degrees of freedom.
 - In almost all situations encountered in practice you can use the usual empirical rule for Normal distributions
- Assumptions:
 - Simple random sample from a large population, that is, independent observations

- No bias in data collection
- Population mean is the parameter you want to estimate

Example 26.3. Continuing with the Ames housing data set. Now suppose we want to compare sale prices of single family homes with other homes (including condos, townhouses, etc). In particular, do single family homes tend to have higher sale prices on average than other homes?

The table below summarizes the sample data.

	count	mean	std	min	25%	50%	75%	max
SingleFamily								
False	505.0	161.511398	60.394023	55.000	120.0	147.4	190.0	392.5
True	2425.0	184.812041	82.821802	12.789	130.0	165.0	220.0	755.0

1. Explain how you could use simulation to approximate the p-value.
2. Use the sample data to conduct by hand an appropriate hypothesis test.
3. Write a clearly worded sentence interpreting the p-value in context.
4. Does the p-value provide evidence that single family homes tend to have higher sale prices on average than other homes? Explain.
5. Does the p-value provide evidence that single family homes tend to have *much* higher sale prices on average than other homes? Explain.

6. How could we address the question in the previous part?

Two-sample t test for a difference in population (or treatment) means.

- Situation: binary explanatory variable (population/treatment 1, 2) and numerical response variable
- Parameter: $\mu_1 - \mu_2$ is the difference in population means (treatment means)
- Goal: assess the strength of the evidence against the “no difference” or “no treatment effect” hypothesis
- The null hypothesis is $H_0 : \mu_1 - \mu_2 = 0$
- The alternative hypothesis is one of $H_a : \mu_1 - \mu_2 > 0$, $H_a : \mu_1 - \mu_2 < 0$, or $H_a : \mu_1 - \mu_2 \neq 0$
 - When in doubt, use a two-sided alternative and a two-tailed p-value.
- Statistic: $\bar{x}_1 - \bar{x}_2$ is the difference in sample means for samples (groups) of size n_1, n_2
- The SE of the difference in sample means is

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{SE^2(\bar{x}_1) + SE^2(\bar{x}_2)}$$

- Test statistic:

$$t = \frac{\text{observed} - \text{hypothesized}}{SE} = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
- The p-value is the probability of observing a value at least as extreme as t , in the tail direction indicated by the alternative hypothesis, for the t -distribution with (complicated) degrees of freedom.
 - In almost all situations encountered in practice you can use the usual empirical rule for Normal distributions
- Assumptions:
 - Either
 - * The data are obtained through two unbiased and independent random samples from large populations

- Or from one unbiased random sample which is classified according to two variables, and the two groups can be considered independent of each other.
- * The data are obtained from an experiment in which subjects were randomly assigned to two treatments.
- No bias in data collection
- Difference in population means is the parameter you want to estimate
- The t procedures for population means are valid as long as the sample size is large enough, regardless of the shape of the population distribution.
- **However, if the population distributions are highly skewed, then the population *medians* might be more appropriate parameters to consider as measures of center.**
- Which is more appropriate, the population means or the population medians, depends on the purpose for the inference.
- Also remember that sample means and SDs, and hence t procedures, are sensitive to outliers and extreme values.

Example 26.4. In Example 26.2 and Example 26.3 we both performed a null hypothesis test and computed a confidence interval. Discuss the advantages and disadvantages of each approach.

- Confidence intervals and hypothesis tests are commonly used in conjunction, but they address different questions
 - Hypothesis test: How strong is the evidence that there is a difference (or association or effect)?
 - Confidence interval: How large is the difference (or association or effect)?
- A confidence interval
 - provides a range of plausible values
 - but the cut-off between plausible and implausible is arbitrary (e.g. 95% confidence corresponds to $\alpha = 0.05$)
 - * But be especially careful about interpreting confidence when conducting multiple confidence intervals.
- A null hypothesis test
 - Considers the plausibility of only a single hypothesized value

- But provides a precise measure — the p-value — of the plausibility of the particular hypothesized value in light of the observed data
 - * But be especially careful about interpreting p-values when conducting multiple hypothesis tests.

27 Some issues with hypothesis testing

- Remember: the smaller the p-value, the stronger the evidence to reject the null hypothesis.
 - But how small is small enough?
- A “significance level”, usually denoted α , is a pre-specified cut-off for how small the p-value must be in order to reject the null hypothesis. The most common value is $\alpha = 0.05$.
 - HOWEVER, in general, do NOT rely too strictly on a significance level α .
 - In fact, for years the [American Statistical Association](#) has said to stop using values like 0.05 to make binary reject/not reject decisions.
- If the p-value is small the result is sometimes referred to as “statistically significant”.
 - HOWEVER, we will see that this is a poor choice of words.
 - In fact, for years the [American Statistical Association](#) has said to stop using the term “statistically significant”. Quote: “‘statistically significant’—don’t say it and don’t use it.”
- Reporting the actual p-value and interpreting it as a sliding scale of weak/moderate/strong evidence to reject the null hypothesis is much more informative than just reporting “reject H_0 at level $\alpha = 0.05$ ” or “p-value < 0.05 ”.
- Remember that hypothesis testing and p-values are just small parts of a much bigger picture. A hypothesis test provides some information, but there are many important questions that hypothesis testing and p-values do not have the ability to address.

Example 27.1. A 2007 paper¹ suggests that in coin-tossing there is a particular “dynamical bias” that causes a coin to be slightly more likely to land the same way up as it started. That is, the paper claims that a coin is more likely to land facing Heads up if it starts facing Heads up than if it starts facing Tails up; likewise if it starts facing Tails up. Is this really true? Two students at Berkeley investigated this phenomenon by tossing a coin many, many times.

1. Define the parameter p in words and state the null and alternative hypotheses using symbols.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

2. Suppose that out of 10000 flips, 5083 landed the same way they started. Based on this data, can you conclude that a coin is more likely to land the same way it started using a strict “significance level” of $\alpha = 0.05$? Compute the p-value and state your conclusion at level $\alpha = 0.05$ in context. Note: One goal of the example is to show you one reason why using a strict α is a bad idea.
3. Repeat the previous part assuming 5082 out of 10000 flips landed the same way they started.
4. Compare the two previous parts. Give one reason why you should be wary of using a “significance level” (like 0.05) too strictly.
5. The students actually tossed the coin 40,000 times!². Of the 40,000 flips, 20,245 landed the same way they started. Compute the p-value and state a conclusion in context. (Do NOT use a strict α level.) Which result—20245 out of 40000 or 5083 out of 10000—gives stronger evidence to reject the null hypothesis?
6. Would you say the results of the real study, in the previous part, are newsworthy? Or important? Or “significant”? Using this problem as an example, explain why “statistically significant” is a poor choice of words. In other words, explain why “statistical significance” is not the same as practical importance.

²If the group sizes are not the same, you would compute a weighted average.

- In general, do NOT rely too strictly on a significance level α .
- Reporting the actual p-value and interpreting it as strong/moderate/weak evidence to reject the null hypothesis is more informative than just reporting “reject H_0 at level $\alpha = 0.05$ ” or “p-value < 0.05 ”.
- Beware the meaning of “significant”.
- A “statistically significant” result is not necessarily practically important or meaningful. Rather, it is just is more extreme than what would be expected to occur just due to chance if the null hypothesis were true, but not necessarily large in practical terms.
- Especially when the sample size is large, even a small observed difference can result in a small p-value.
- On the other hand, an observed result might have practical significance even when the p-value is not small.
- A confidence interval estimates the size of the difference, which helps in judging practical significance. It is also important to consider *effect size*.
- Do NOT use the terminology “statistically significant”.

Example 27.2. (For this exercise, you’ll need to believe it’s possible for a person to have ESP.) “Ganzfeld studies” test psychic ability (like ESP). Ganzfeld studies involve two people, a “sender” and a “receiver”, who are placed in separate acoustically-shielded rooms. The sender looks at a “target” image on a television screen (which may be a static photograph or a short movie segment playing repeatedly) and attempts to *transmit with their mind* information about the target to the receiver. The receiver is then shown four possible choices of targets, one of which is the correct target and the other three are “decoys.” The receiver must choose the one transmitted by the sender³.

1. We will now conduct some Ganzfeld tests to see if anyone in the class has ESP.
 - Find a partner. Identify one person as the sender and one as the receiver.
 - The sender rolls a four-sided die, looks at it (without letting the receiver see), and then tries to transmit the number rolled to the mind of the receiver.
 - After receiving the mental image, the receiver identifies the number. Record whether the answer is correct or not.
 - Repeat the above process for a total of 9 trials with the same receiver. The receiver should record the number of trials (out of 9) they got correct.
 - Then switch sender/receiver roles and do it again. Each receiver should record the number of trials (out of 9) that they got correct.

³This number assumes independence of confidence intervals — $0.95^{10} \approx 0.60$ — but similar ideas apply more generally.

2. Use your results to conduct a hypothesis test to see if you have evidence that you did better than just guessing (and so you have some kind of ESP!)

3. Suppose we had set a cut-off of 0.05. Based on this strict cutoff, do you have enough evidence to reject the null hypothesis? That is, based on this cutoff, is your p-value small enough to conclude that you have ESP? (Hint: if you got at least 5 out of 9 correct your answer should be “yes”.)

4. Consider a student who rejects the null hypothesis. Could their conclusion be wrong? How?

5. Consider a student who fails to reject the null hypothesis. Could their conclusion be wrong? How? (Again, you’ll need to believe it’s possible for a person to have ESP.)

6. Using the cutoff of 0.05, did any receiver in our class (of about 30 students) discover evidence that they have ESP? Should we be surprised that this happened?

- The null hypothesis H_0 is “innocent until ‘proven’ guilty”. That is, the null hypothesis is not rejected unless there is convincing evidence against it.
- When making a conclusion — reject H_0 , or fail to reject H_0 — there are two errors that could be made:
 - **Type I error:** Reject H_0 when H_0 is actually true (unknownst to the researcher)
 - * “False alarm” or “false discovery”
 - * Analogous to sending an innocent person to jail.
 - **Type II error:** Fail to reject H_0 when H_0 is actually false (unknownst to the researcher)
 - * “Missed opportunity”
 - * Analogous to letting a guilty person go free.
- The probability of making a Type I error can be managed by setting a “significance” level α .
- If decisions are made according to the rule “reject the null hypothesis only when p-value $< \alpha$ ” then the probability of making a Type I error is (at most) α .
- But there is a tradeoff: decreasing the probability of making a Type I error will increase probability of making a Type II error.
- The **power** of a test is the probability of correctly rejecting the null hypothesis when the the null hypothesis is false.
 - The power is the probability of making a “true discovery”.

		What is true (unknownst to the jury)	
		Null hypothesis is true (defendant is innocent)	Null hypothesis is false (defendant is guilty)
What jury decides (based on evidence)	Reject null hypothesis (Jury finds defendant guilty)	Type I error (false alarm)	Correct Decision
	Do not reject null hypothesis (Jury finds defendant not guilty)	Correct Decision	Type II error (missed opportunity)

- Hypothesis tests are appropriate when there is a clearly defined research question and conjecture (i.e. alternative hypothesis), and sample data is collected for this purpose.
- If many hypothesis tests are performed, eventually you will reject a null hypothesis — just by chance — even if all the null hypotheses are really true.
 - For example, in a set of 35 tests all conducted using strict level $\alpha = 0.05$, there is about an 84 percent chance of rejecting at least one of the null hypotheses when all 35 null hypotheses are true.

- Beware of **p-hacking**, which involves repeating hypothesis tests until a small p-value is found
- The issue of multiple testing/p-hacking is another reason why you should not rely too strictly on a “significance” level α .
- Do NOT use a strict level α to make a strict reject H_0 /fail to reject H_0 conclusion based on observed sample data about a particular hypothesized value or research question
 - Rather, report the p-value and assess strength of evidence in favor of rejecting the null hypothesis on a sliding scale.
 - But remember the p-value doesn’t tell you anything about how big or meaningful the difference is—but rather just if it’s not due to random chance—especially if the sample size is large
 - Be especially careful about using α or interpreting p-values when conducting multiple hypothesis tests on the same data. Beware p-hacking.
- Do USE “significance” levels
 - For research planning purposes — to manage probability of Type I error, to investigate power, and to determine an appropriate sample size — *before* collecting data
 - When determining a *range* of plausible values for the parameter, e.g., to determine the cutoff between plausible/not plausible in a confidence interval.
 - * A 95% confidence is essentially the range of hypothesized values that would not be rejected at level 0.05 based on the observed data.
- Remember that hypothesis testing and p-values are just small parts of a much bigger picture. A hypothesis test provides some information, but there are many important questions that hypothesis testing and p-values do not have the ability to address.

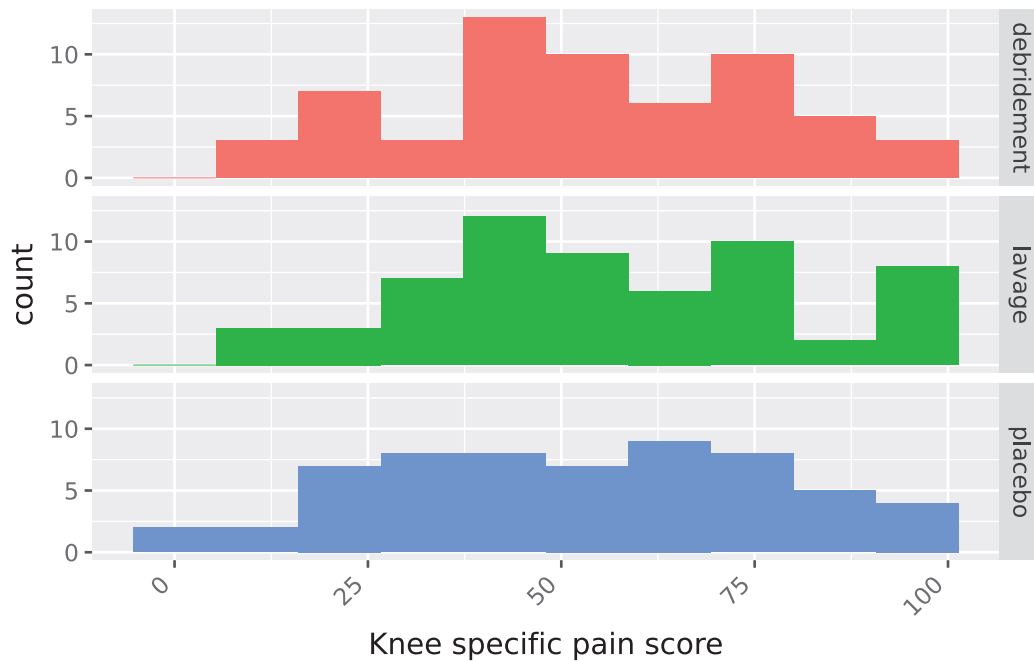
28 Comparing multiple means: ANOVA F procedures

Example 28.1. When medical therapy fails to relieve the pain of osteoarthritis of the knee, arthroscopic lavage or debridement is often recommended. More than 650,000 such procedures are performed each year at a cost of roughly \$5,000 each. A recent double blind experiment investigated the effectiveness of these procedures¹. 180 subjects with osteoarthritis of the knee were randomly assigned to receive one of three treatments: arthroscopic lavage (which involved flushing the knee joint with fluid), arthroscopic debridement (which involved both lavage and shaving of cartilage and removal of debris), and placebo surgery (which involved incisions in the knee, but no insertion of instruments). Various outcomes were measured; the summaries below summarize scores two years after the procedure on a knee-specific pain scale (0-100, with higher scores indicating more severe pain). Does any procedure tend to result in better pain scores on average?

1. State the null and alternative hypotheses (in words is fine).
2. The summarizes the sample data. Do you think there will be any evidence to reject the null hypothesis?
3. Suggest a statistic which could be used to perform a hypothesis test.

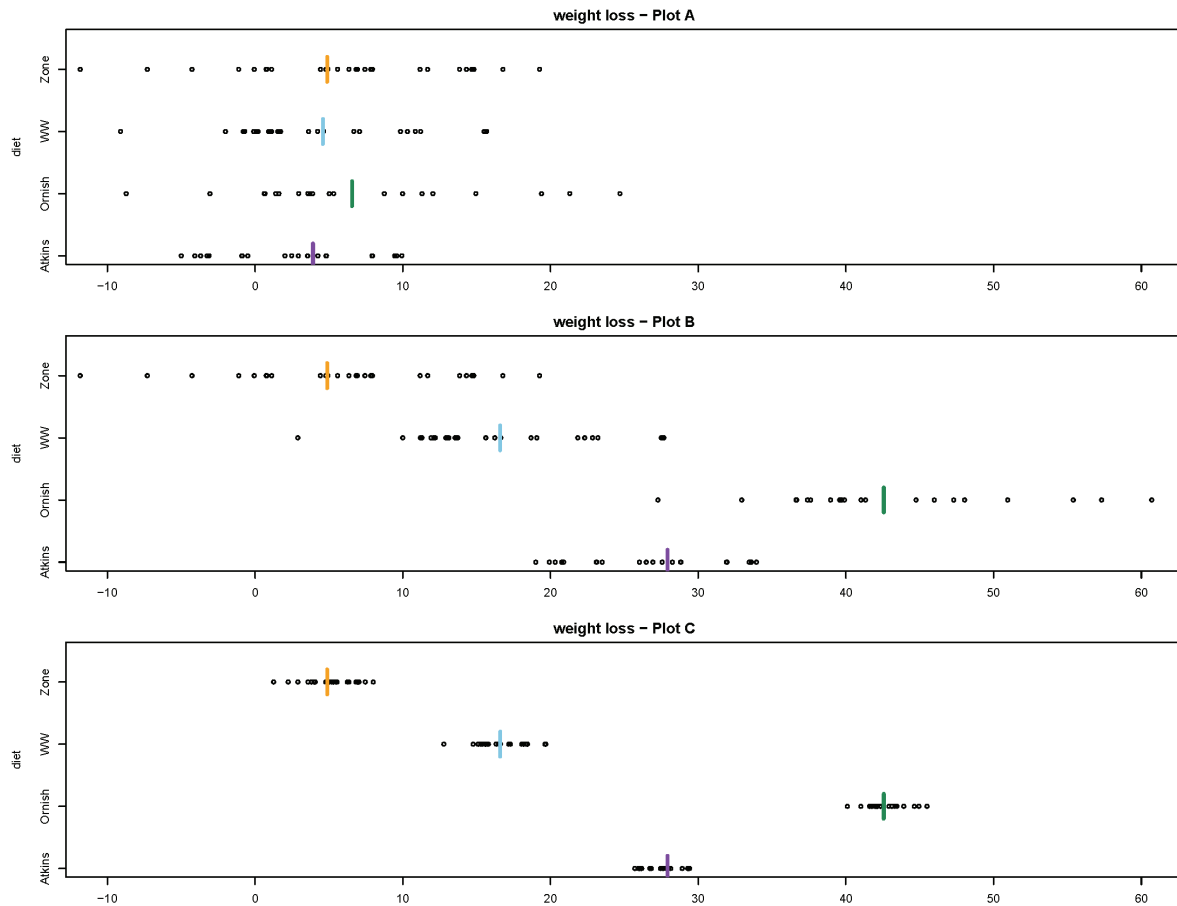
¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

	count	mean	std	min	25%	50%	75%	max
treatment								
debridement	60.0	53.983333	23.304536	10.0	40.75	52.0	76.25	97.0
lavage	60.0	56.733333	24.150564	8.0	41.00	53.5	73.50	99.0
placebo	60.0	52.450000	25.079146	3.0	31.00	54.0	70.25	97.0



Example 28.2. Consider the following plots, which are all on the same scale. (This example just illustrates some general ideas so the context isn't important, but the data relate to a study on different types of diet plans (Atkins, etc) and weight loss.) Plot A displays the study data. Plots B and C show hypothetical data sets with the same group sizes as in Plot A. Group means are indicated by the colored vertical lines.

- The SD for each group in Plot B equals the SD for the corresponding group in Plot A.
- The mean for each group in Plot C equals the mean for the corresponding group in Plot B.



1. Rank the plots in order from weakest to strongest evidence to reject the null hypothesis of no difference in means between treatments. That is, rank the plots in order from largest to smallest p-value.
2. Compare plots A and B.
 - The variability *between* groups in Plot B is (**choose one:** $>$, $<$, $=$) the variability *between* groups in Plot A.
 - The variability *within* groups in Plot B is (**choose one:** $>$, $<$, $=$) the variability *within* groups in Plot A.
3. Compare plots B and C.
 - The variability *between* groups in Plot C is (**choose one:** $>$, $<$, $=$) the variability *between* groups in Plot B.
 - The variability *within* groups in Plot C is (**choose one:** $>$, $<$, $=$) the variability *within* groups in Plot B.

4. Rank the plots in order from smallest to largest F statistic.

- Example 28.2 illustrates the idea behind the F statistic: we reject the null hypothesis of no difference in means between groups/populations/treatments if the observed variability *between* groups is large relative to the observed variability *within* groups.
- Roughly, we reject the null hypothesis if the ratio

$$\frac{\text{variability between groups}}{\text{variability within groups}}$$

is large enough. This idea is known as **analysis of variance (ANOVA)**.

- The name ANOVA refers to *what* we are doing, and not *why* we are doing it.
- Remember, we are analyzing the variance to make a conclusion about the *means*
- The F statistic quantifies the above ratio. The larger the value of the F statistic the stronger the evidence to reject the null hypothesis (of no difference) in favor of the alternative hypothesis (of some difference). That is, the larger the F statistic the smaller the p-value.

Example 28.3. Continuing Example 28.1. The computation of the F statistic is full of technical details. We will only do a rough hand computation which illustrates the main ideas, but we'll usually rely on software to compute it. The computation is somewhat simplified when the sample size is the same for each explanatory variable group.

1. What statistic can be used to measure the variability *within* the Debridement group? Lavage? Placebo? How could you combine these numbers to measure, in a single number, *variability within groups*?
2. If the null hypothesis is true, what would you expect about the group means? What statistic can be used to measure the variability *between* groups?

3. Regardless of whether or not the null hypothesis is true, which of the values from the previous parts would you expect to be larger? Why?
4. So far, we have used the group means and group SDs in our calculations. What other feature of the groups do you think will influence the results? How?
5. Compute the F statistic.
6. What value would you expect F to be if the null hypothesis is true? Specify in detail how you could use simulation to approximate the null distribution of the F statistic.
7. Figure 28.1 displays results from an applet used to run 10000 repetitions of the simulation. How could you use the simulation results to approximate the p-value? Could you use the empirical rule for Normal distributions?
8. State a conclusion in context. Are the results “significant”?
9. If we were to compute a series of pairwise confidence intervals for the difference in group means — Debridement — Lavage, Debridement — Placebo, Lavage — Placebo — what value would each of the confidence intervals contain? Why?

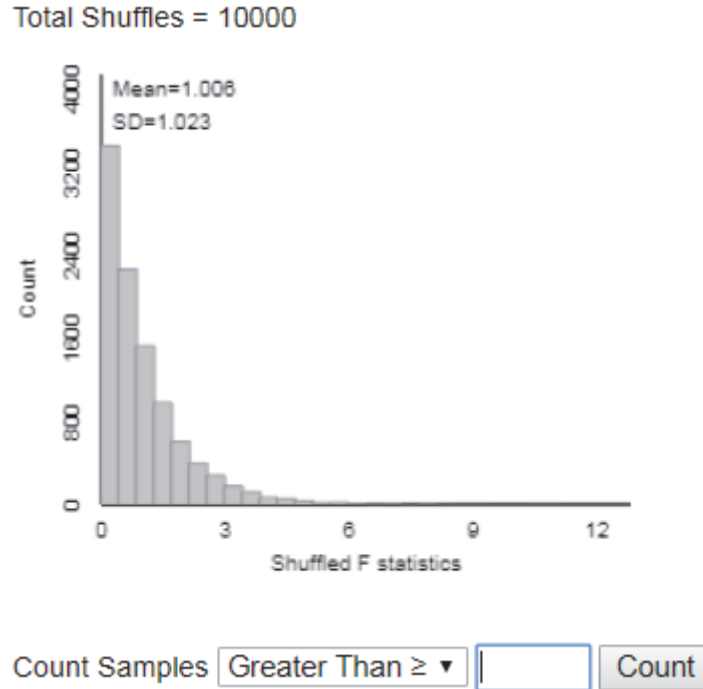


Figure 28.1: Permutation null distribution of F statistic for Example 28.1.

Hypothesis test for comparing multiple means: “One-way ANOVA F test”

- Situation: N observations of numerical response variable; categorical explanatory variable with g groups
- Parameters: μ_j 's are the population/process/treatment response means; σ is the common group SD of responses
- Goal: test for any differences in means
- The null hypothesis is $H_0 : \mu_1 = \dots = \mu_g$ (“no difference” or “no treatment effect”)
- The alternative hypothesis is not H_0 ; that is, there is some difference in means between groups, $H_1 : \mu_j \neq \mu_k$ for some $j \neq k$
- Test statistic: one-way ANOVA F statistic

$$F = \frac{\sum_{j=1}^g n_j (\bar{x}_j - \bar{x})^2 / (g - 1)}{\sum_{j=1}^g (n_j - 1) s_j^2 / (N - g)}$$

- The p-value is the probability of observing a value at least as large as the observed F based on an F distribution with $g - 1$ degrees of freedom in the numerator and $N - g$ degrees of freedom in the denominator
- Assumptions:
 - Either:
 - * The data are obtained through multiple unbiased and independent random samples from large populations
 - Or from one unbiased random sample which can be classified into multiple groups, and the groups can be considered independent of each other.
 - * The data are obtained from an experiment in which subjects were to two treatments.
 - No bias in data collection
 - Comparing *means* is an appropriate analysis
 - Equal variance: True SD of responses is same, σ , for each population/process/treatment.
 - * Typically satisfied through random assignment.
 - * Might not be satisfied if the data are sampled from different populations/processes.
 - * One rule of thumb is that the largest observed sample SD should be no more than twice the smallest.
- Special case: when the group size n is the same for all g groups is

$$\begin{aligned}
 F &= \frac{\text{group size} \times \text{variance of group means}}{\text{mean of group variances}} \\
 &= \frac{n \sum_{j=1}^g (\bar{x}_j - \bar{x})^2 / (g - 1)}{\sum_{j=1}^g s_j^2 / g}
 \end{aligned}$$

- The calculations are traditionally summarized in an **ANOVA table**

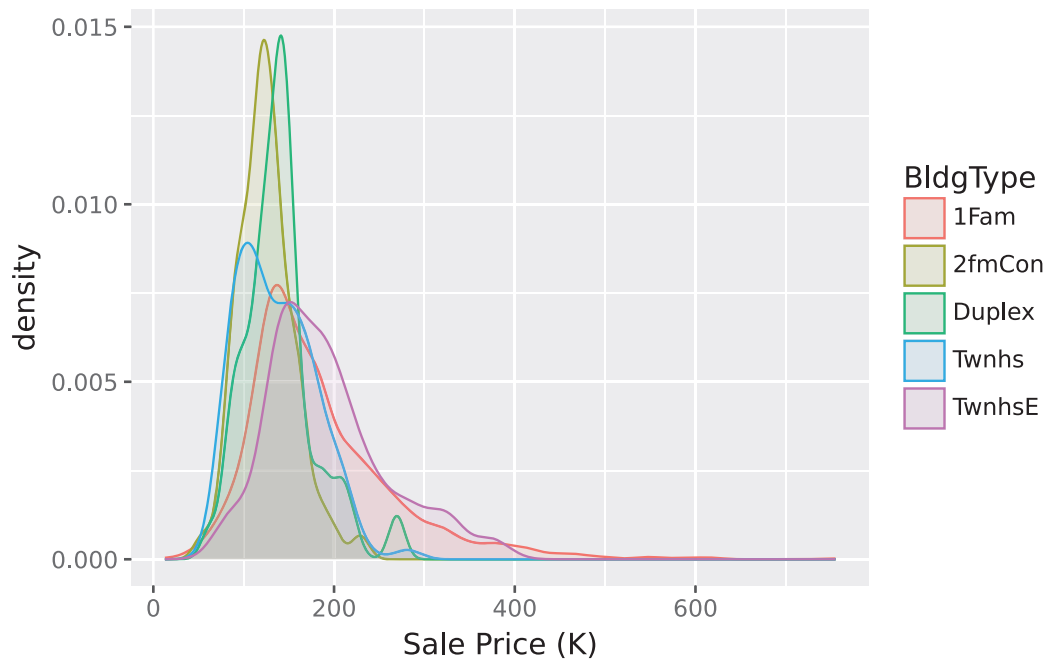
Source	df	SS (Sum of squares)	MS (Mean Square)	F
Between groups (a.k.a. “treatment”)	$g - 1$	$SS(\text{between}) = \sum_{j=1}^g n_j (\bar{x}_j - \bar{x})^2$	$\frac{SS(\text{between})}{g - 1}$	$\frac{MS(\text{between})}{MS(\text{within})}$
Within groups (a.k.a. “error”)	$N - g$	$SS(\text{within}) = \sum_{j=1}^g (n_j - 1) s_j^2$	$\frac{SS(\text{within})}{N - g}$	
Total	$N - 1$	$(SS(\text{between}) + SS(\text{within})) = \sum_{i=1}^N (x_i - \bar{x})^2$		

Table 28.4: ANOVA table for Example 28.4.

	sum_sq	df	F	PR(>F)
C(BldgType)	6.454111e+05	4.0	26.151358	2.475923e-21
Residual	1.804713e+07	2925.0	NaN	NaN

Example 28.4. Continuing with the Ames housing data set. Now suppose we want to compare sale price of homes by building type: single-family (1Fam), two-family (2fmCon), duplex, townhouse end unit (TwnhsE), townhouse inside unit (Twnhs).

	count	mean	std	min	25%	50%	75%	max
BldgType								
1Fam	2425.0	184.812041	82.821802	12.789	130.0000	165.000	220.000	755.00
2fmCon	62.0	125.581710	31.089240	55.000	106.5625	122.250	140.000	228.95
Duplex	109.0	139.808936	39.498974	61.500	118.8580	136.905	153.337	269.50
Twnhs	101.0	135.934059	41.938931	73.000	100.5000	130.000	170.000	280.75
TwnhsE	233.0	192.311914	66.191738	71.000	145.0000	180.000	222.000	392.50

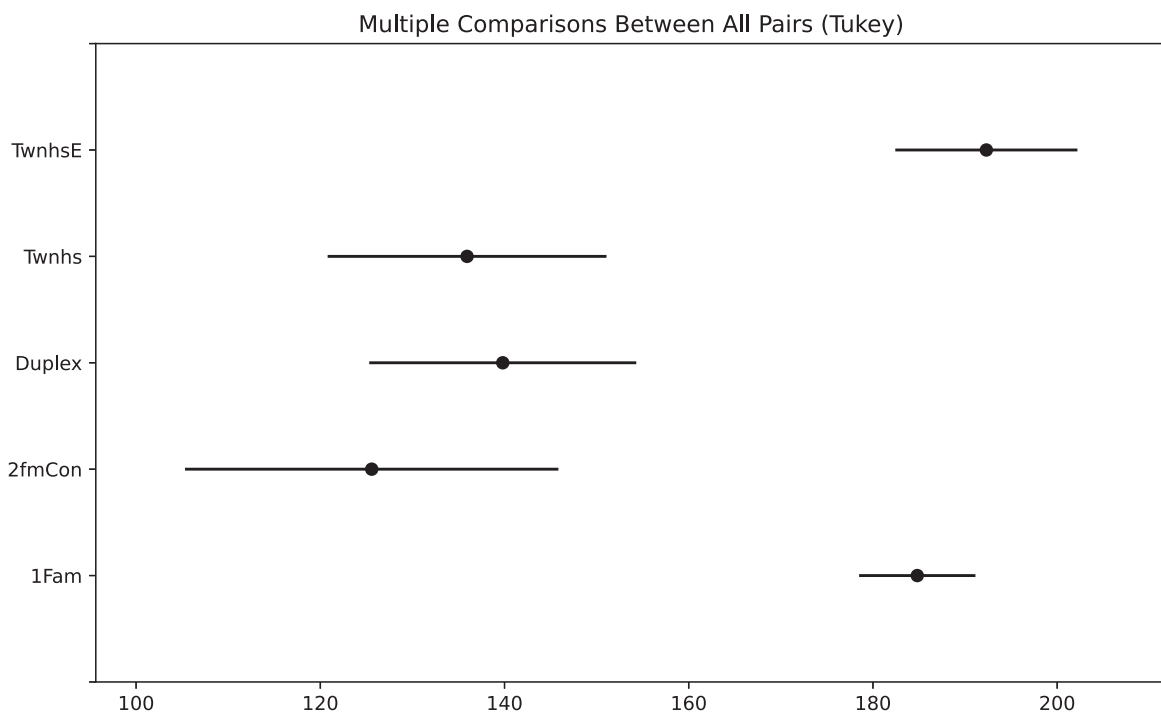
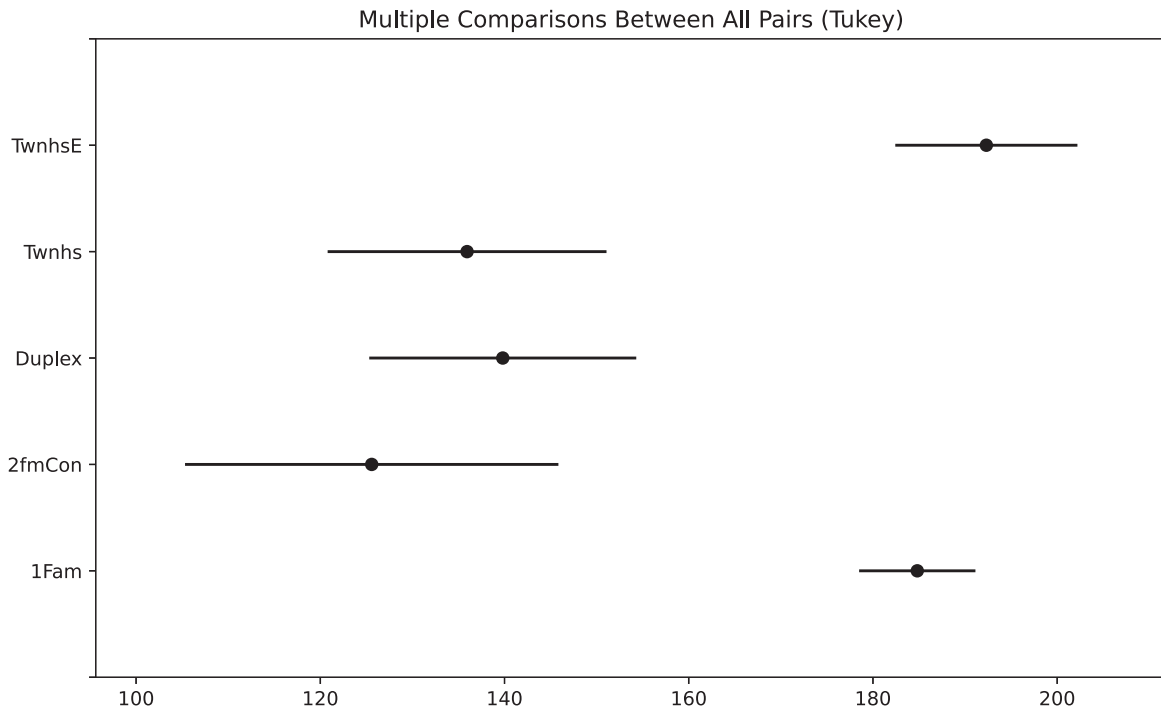


1. Table 28.4 is the ANOVA table. State the conclusion of the ANOVA F test in context.

Table 28.5: Tukey pairwise 95% confidence intervals for Example 28.4.

Multiple Comparison of Means - Tukey HSD, FWER=0.05						
group1	group2	meandiff	p-adj	lower	upper	reject
1Fam	2fmCon	-59.2303	0.0	-86.8048	-31.6559	True
1Fam	Duplex	-45.0031	0.0	-65.9952	-24.0111	True
1Fam	Twnhs	-48.878	0.0	-70.6511	-27.1049	True
1Fam	TwnhsE	7.4999	0.6328	-7.2051	22.2048	False
2fmCon	Duplex	14.2272	0.786	-19.8771	48.3316	False
2fmCon	Twnhs	10.3523	0.9255	-24.2382	44.9429	False
2fmCon	TwnhsE	66.7302	0.0	36.0924	97.368	True
Duplex	Twnhs	-3.8749	0.9965	-33.4861	25.7363	False
Duplex	TwnhsE	52.503	0.0	27.6234	77.3825	True
Twnhs	TwnhsE	56.3779	0.0	30.8358	81.9199	True

2. Consider the conclusion of the F test. What are some natural follow-up questions? How might you use the data to answer them?
3. Table 28.5 contains the results of Tukey pairwise 95% confidence intervals. Interpret these intervals in context.



- Rejecting the null hypothesis based on an F statistic only tells you that “the mean for

at least one of the treatments (populations) is not the same as the mean for at least one of the other treatments (populations)”.

- The natural follow-up questions are: which treatment (population) means differ? and by how much?
- One approach when you have rejected the null hypothesis is to calculate — using software — all pairwise confidence intervals. In particular, which intervals do not contain zero?
- When computing multiple confidence intervals, the multiple (z^* or t^*) should be increased² to account for the **issue of multiple comparisons** (recall Example 24.5 and related notes).
- **Tukey’s method** is a widely used method for constructing simultaneous confidence intervals for *all pairwise comparisons of means*. The confidence intervals computed using Tukey’s method already contain an adjustment to account for multiple comparisons.

²There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.